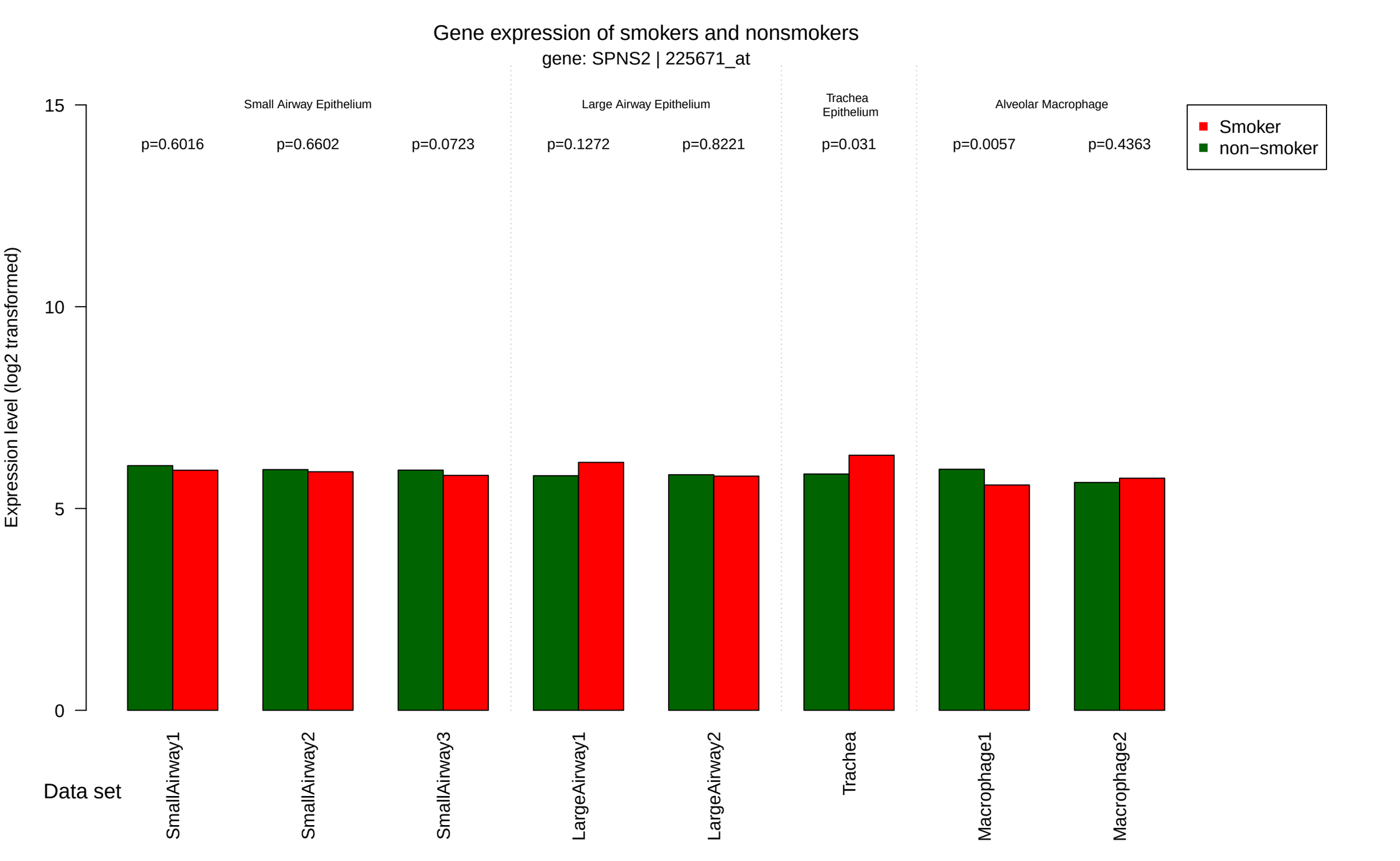
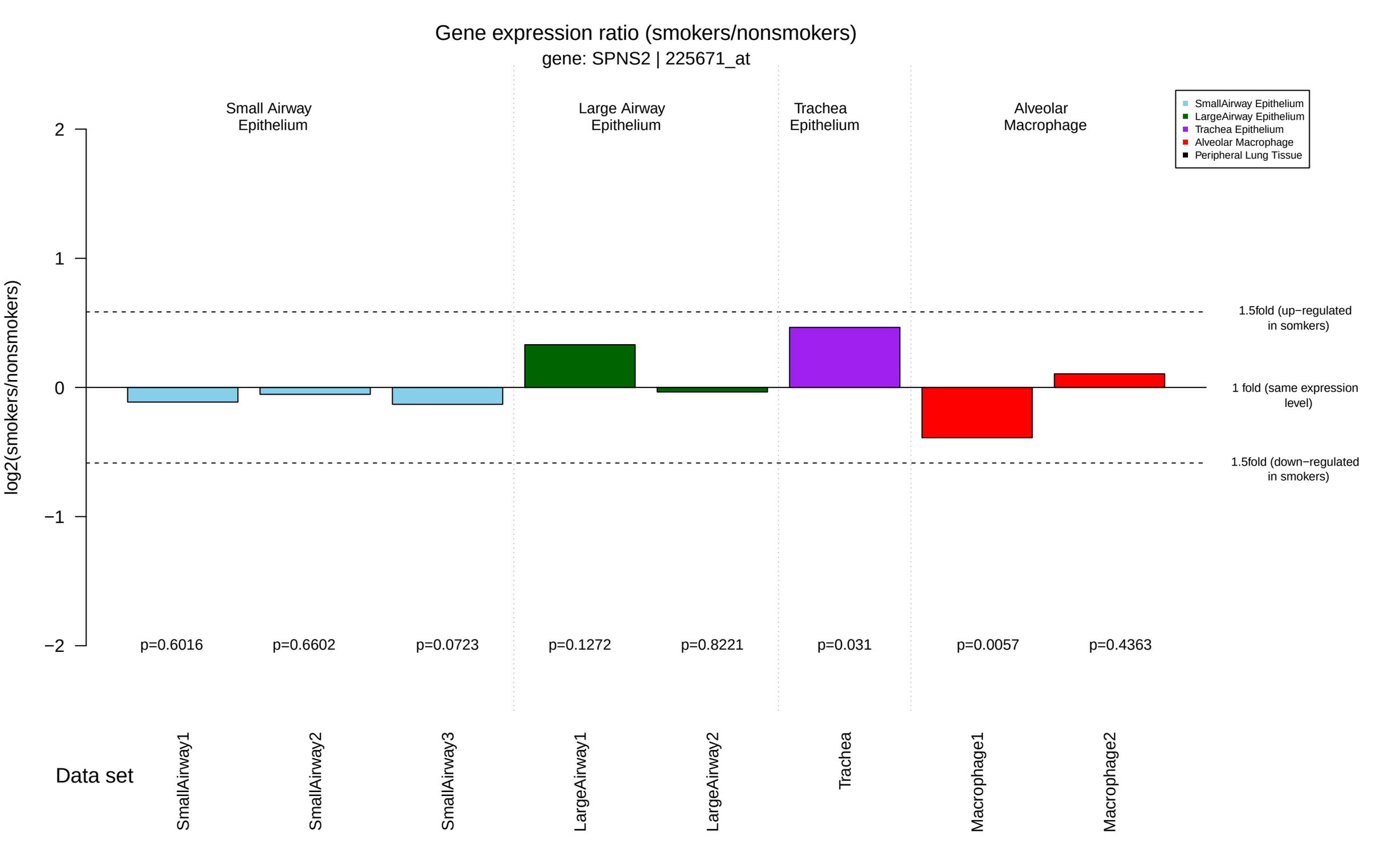
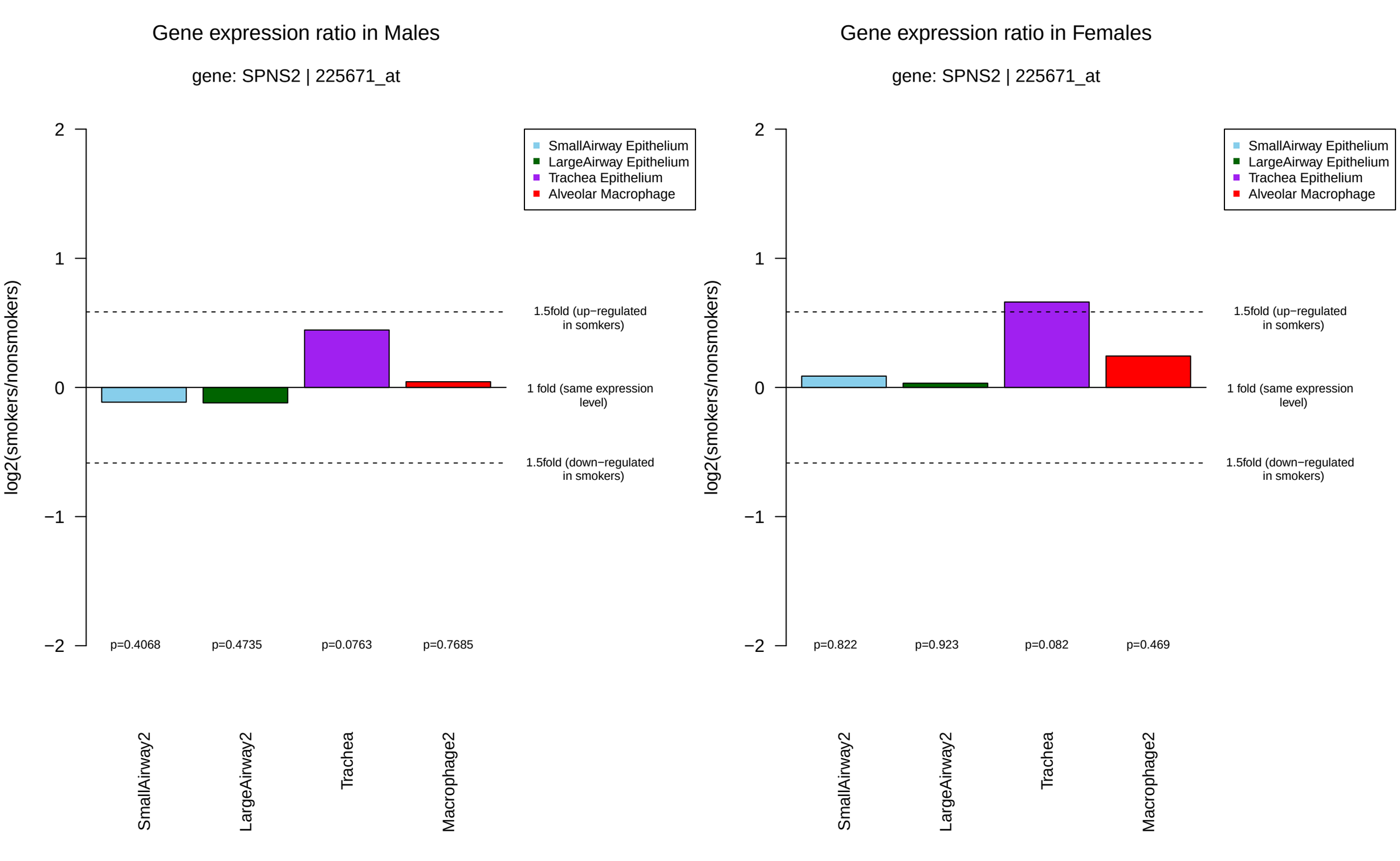
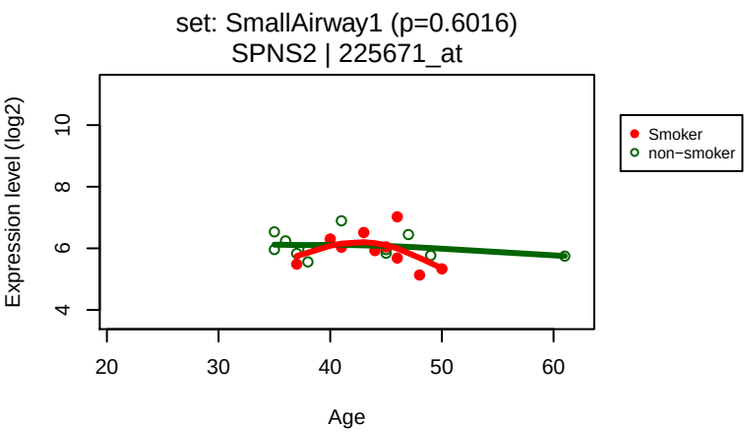
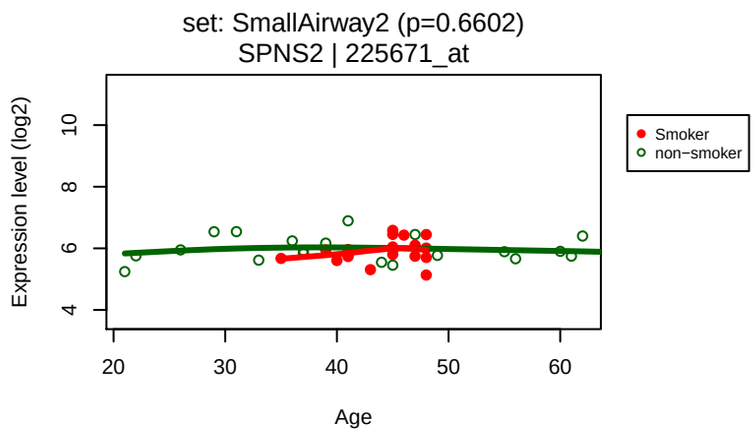
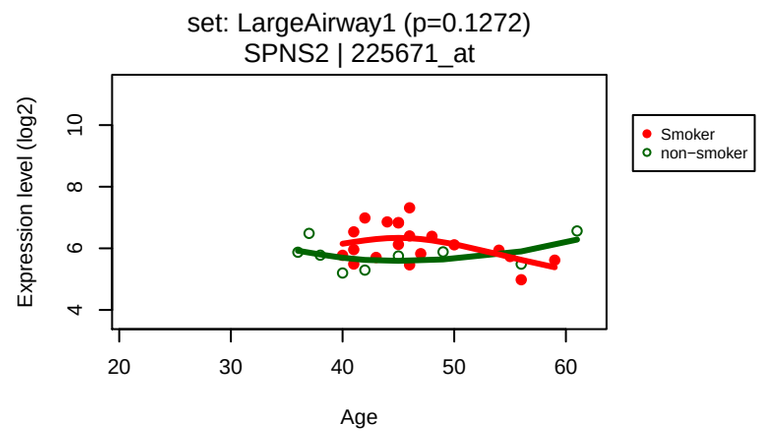

Gene expression of smokers and nonsmokers
gene: SPNS2 | 225671_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6016
p=0.6602
p=0.0723
p=0.1272
p=0.8221
p=0.031
p=0.0057
p=0.4363
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

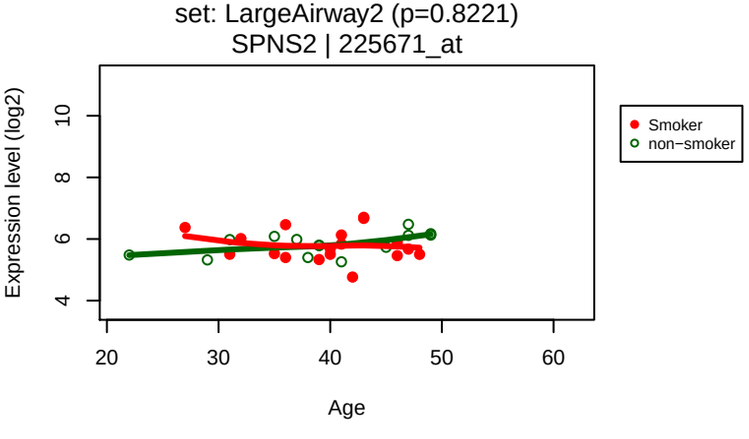

Gene expression ratio (smokers/nonsmokers)
gene: SPNS2 | 225671_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6016
p=0.6602
p=0.0723
p=0.1272
p=0.8221
p=0.031
p=0.0057
p=0.4363
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

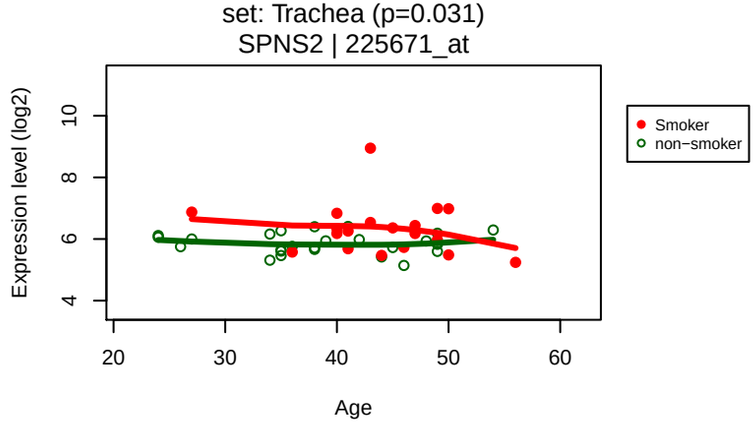

Gene expression ratio in Males
gene: SPNS2 | 225671_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4068
p=0.4735
p=0.0763
p=0.7685
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPNS2 | 225671_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.822
p=0.923
p=0.082
p=0.469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

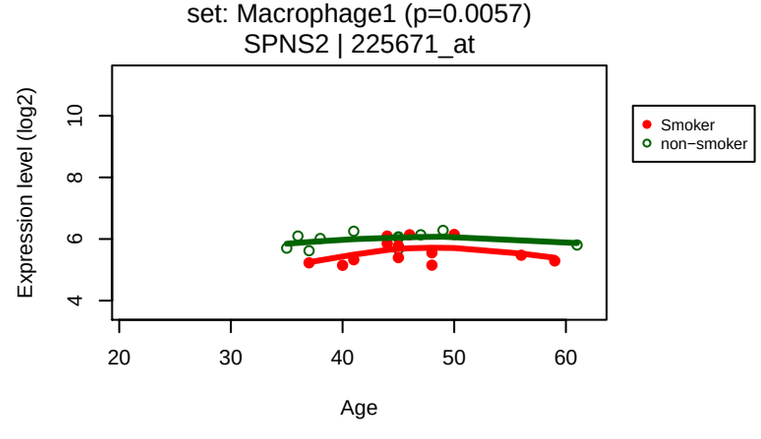

set: SmallAirway1 (p=0.6016)
SPNS2 | 225671_at
Expression level (log2)
Age
Smoker
non−smoker

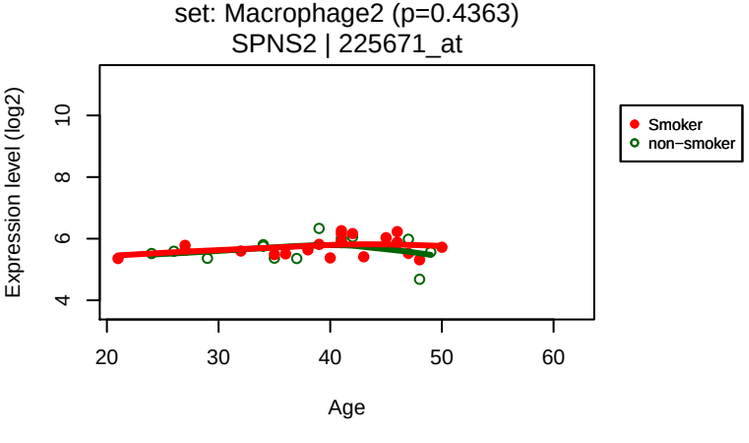

set: SmallAirway2 (p=0.6602)
SPNS2 | 225671_at
Expression level (log2)
Age
Smoker
non−smoker

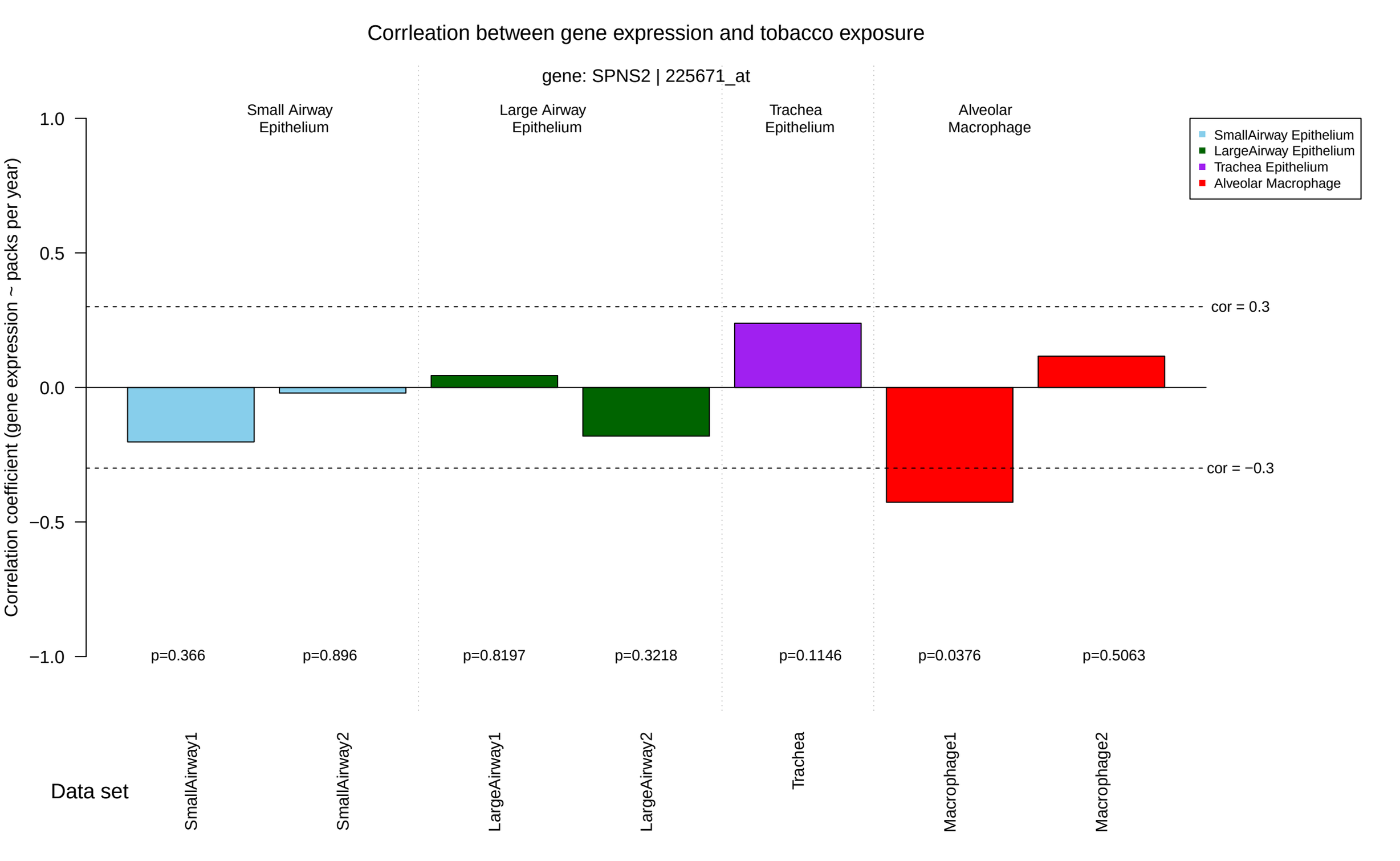

set: LargeAirway1 (p=0.1272)
SPNS2 | 225671_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8221)
SPNS2 | 225671_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.031)
SPNS2 | 225671_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0057)
SPNS2 | 225671_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4363)
SPNS2 | 225671_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPNS2 | 225671_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.366
p=0.896
p=0.8197
p=0.3218
p=0.1146
p=0.0376
p=0.5063
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2